

# Calculus 1 and 2

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## 1 Introduction

Calculus is the starting place for mathematics for many engineers in college, but I hesitate to call it the foundation. In truth, our mathematical foundation takes its roots as far back as high school algebra or even arithmetic, but it is always calculus that is given the focus.

Why am I talking about algebra in a calculus review? Because often times our struggles with performing calculus come from confusion in algebraic operations. This section of notes covers the calculus itself, so if this is not helpful and the video lectures accompanying the calculus review were lacking, I recommend looking into algebraic operations.

Moving on to calculus, let's ask ourselves a question. What is Calculus? What are its aims? What can it do? Calculus is the math of change. It allows us to look at the behavior of algebraic functions. It aims to inform on the properties of mappings. Thus it allows us to build models of changing systems, or even lend depth to models of static systems.

How does calculus accomplish this? Calculus fundamentally draws boundary around points of mapping and says this is what is approximately happening at this point. More, calculus takes smaller and smaller open sets centered at a point to investigate the properties of the function around that point. This is done with limits. So let's kick off our investigation with what a limit even is!

## 2 Definitions and Theorems

We are going to jump right in with the epsilon delta definition of a limit as an understanding of what this is saying and doing is the best way to truly understand what a limit is.

Why is this not taught this way in calculus then? Because in calculus 1 we haven't yet learned the language of mathematics. Fortunately, we have! That was the notation we considered in the notation notes and introduction video.

### Epsilon Delta definition of a Limit

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  and let  $x, c, L \in \mathbb{R}$ . Then we say the limit of  $f(x)$  as  $x$  approaches  $c$  is  $L$  and is written as  $x \rightarrow c \implies f(x) \rightarrow L$  or

$$\lim_{x \rightarrow c} f(x) = L$$

if and only if  $\forall \varepsilon > 0, \exists \delta > 0$  such that  $|f(x) - L| < \varepsilon$  whenever  $0 < |x - c| < \delta$ .

So whats that bottom sentence with all the symbols saying? Well its saying that if we draw a boundary of size  $\varepsilon$  along the y-axis and a boundary of size  $\delta$  along the x-axis centered at  $c$  the function  $f$  is going to be inside this boundary.

For this definition of a derivative, our understanding of a continuous function being a function we can draw without picking up our pencil will suffice. However, for those who are interested, after defining a derivative I will more rigorously define continuous.

### Limit Definition of a Derivatives

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a continuous function. Then the derivative of  $f$  with respect to  $x$  is

$$f'(x) = \frac{d}{dx} f(x) = \lim_{x \rightarrow h} \frac{f(x+h) - f(x)}{h}$$

or

$$\frac{d}{dx} f(x) = \lim_{a \rightarrow b} \frac{f(a) - f(b)}{a - b}$$

Notice that these two forms are really the same. In the first form we have simply substituted  $a = b + h$ .

So why present the two forms? The first form is easier to perform computations with, as it has chose a path in which our secant lines can approach a tangent line, where as our second form has not made a decision. This means that we can approach the point of interest from both sides at any varying rate we chose.

Lets now take a more rigorous look at the term continuous for those who are interested.

### Continuous Functions

Suppose  $f : \mathbb{R} \rightarrow \mathbb{R}$  and  $c \in \mathbb{R}$ . Then a function is said to be continuous at the point  $c$  if  $\forall \varepsilon > 0, \exists \delta > 0$  such that  $|f(x) - f(c)| < \varepsilon$  whenever  $|x - c| < \delta$  and  $x \in \mathbb{R}$

Notice the similarity between this definition and the epsilon delta definition of a limit. The key difference being in

the inequality involving  $\delta$ . In our definition of a limit we restrict this inequality to be greater than 0 where as with our definition of a continuous function we allow for our difference to go to zero. This forces  $x \neq c$  in a limit, and forces  $x = c$  to be considered in continuous functions. More broadly, this is what gives us the ability to say a continuous function is a function that can be drawn without picking up our pencils.

### Sequences

An infinite sequence  $\{a_n\}$  is an ordered list of elements of the form

$$a_0, a_1, a_2, \dots$$

### Series

Let  $(a_n)$  be a sequence. We define the series as the sum of elements in  $(a_n)$ , and write

$$\sum_{n=1}^{\infty} a_n$$

Notice the key difference between sequences and series is that sequences are just lists of elements, where as a series sums those elements.

### Riemann Integral

Suppose  $f : \mathbb{R} \rightarrow \mathbb{R}$  is a continuous function on the interval  $[a, b]$ . Then the integral from  $a$  to  $b$  is the area under the curve  $f$  on  $[a, b]$  and is the limit

$$\begin{aligned} \int_a^b f(x)dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x_i \\ &= \lim_{\Delta x_i \rightarrow 0} \sum_{i=1}^n f(x_i) \Delta x_i \end{aligned}$$

In this definition we are summing rectangles with width  $\Delta x_i$  and hight  $f(x_i)$ .

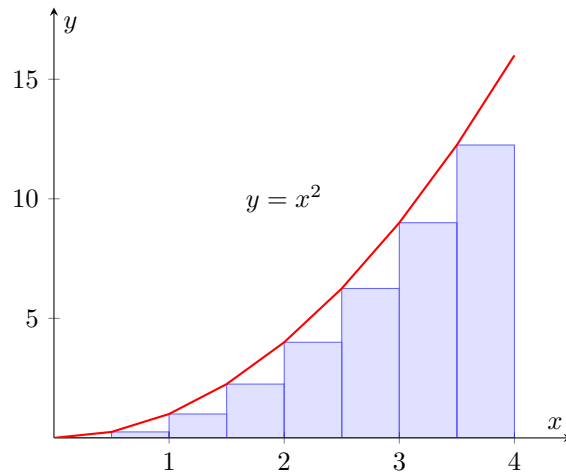


Figure 1: Left Riemann sum approximation with  $n = 8$  subintervals.

The figure above depicts the finite Riemann sum. An integral is going to see the width of each of those rectangles get thinner and thinner while adding more and more rectangles to the interval.

### 3 Examples

So, limits do be existing sometimes. Lets look one. Consider the function  $f : \mathbb{R} \rightarrow \mathbb{R}$  where  $f(x) = \frac{x^2 - 3x - 10}{x^2 - 4}$ . Lets take the limit as  $x \rightarrow -2$ . So we have

$$\begin{aligned} L &= \lim_{x \rightarrow -2} f(x) \\ &= \lim_{x \rightarrow -2} \frac{x^2 - 3x - 10}{x^2 - 4} \end{aligned}$$

We are going to attempt direct substitution, but if we start here we'll have to divide by zero. To avoid this lets attempt to factor.

$$\begin{aligned}
L &= \lim_{x \rightarrow -2} \frac{x^2 - 3x - 10}{x^2 - 4} \\
&= \lim_{x \rightarrow -2} \frac{(x+2)(x-5)}{(x-2)(x+2)} \\
&= \lim_{x \rightarrow -2} \frac{(x+2)(x-5)}{(x+2)(x-2)} \\
&= \frac{(-2-5)}{(-2-2)} \\
&= \frac{(-7)}{(-4)} \\
L &= \frac{7}{4}
\end{aligned}$$

So we have

$$L = \lim_{x \rightarrow -2} f(x) = \lim_{x \rightarrow -2} \frac{x^2 - 3x - 10}{x^2 - 4} = \frac{7}{4}$$

Lets practice a derivative now. We will do the age old example of  $x^2$ .

$$\begin{aligned}
\frac{d}{dx}x^2 &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\
&= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\
&= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\
&= \lim_{h \rightarrow 0} 2x + h \\
&= 2x + 0 \\
\frac{d}{dx}x^2 &= 2x
\end{aligned}$$

Now lets apply the power rule. For reference see the identities document.

$$\frac{d}{dx}x^n = nx^{n-1}$$

So

$$\begin{aligned}\frac{d}{dx}x^2 &= 2x^{2-1} \\ &= 2x^1 \\ &= 2x\end{aligned}$$

So as we can see its the same result. Which is expected given the rules of derivation come primarily from the epsilon delta definition of a derivative which can be found on wikipedia.

Moving to look at sequences, lets consider a classic, the Fibonacci sequence. The Fibonacci sequence is famous for its relation to geometry and nature. The sequence is written  $\{1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, \dots\}$  as you can see each term is the sum of the previous two terms, written as

$$F_n = F_{n-1} + F_{n-2}$$

Another way to write the Fibonacci sequence is as a function of the nth term. This is of the form

$$F_n = \frac{1}{\sqrt{5}} \left[ \left( \frac{1 + \sqrt{5}}{2} \right)^n - \left( \frac{1 - \sqrt{5}}{2} \right)^n \right]$$

As you can see this is simply a function of n. The connection between these two forms is through the golden ratio. I recommend looking into wikipedia to learn more if you are interested.

A few key qualities to observe are a sequences nth term can be written in terms of one or more of its previous terms. Or it can be written explicitly only in terms of n. And most importantly it does NOT sum the terms!

Another quick example of a sequence is the alternating sequence. This has elements and form

$$\begin{aligned}\{1, -1, 1, -1, 1, -1, \dots\} \\ s_n = (-1)^n, \text{ where } n \geq 0\end{aligned}$$

Moving to look at series. First note that an infinite series is the sum of the terms in a sequence. In order to avoid discussing convergence we are going to look at an example that is guaranteed to converge. This type of series is called a telescoping series. The name will become clear with the example.

The example in question we will consider is the series built off the sequence

$$a_n = \frac{1}{n+2} - \frac{1}{n+1}$$

forming the series

$$\begin{aligned} S &= \sum_{n=0}^{\infty} a_n \\ &= \sum_{n=0}^{\infty} \frac{1}{n+2} - \frac{1}{n+1} \end{aligned}$$

Notice that if we take the Nth partial sum

$$S_N = \sum_{n=0}^N \frac{1}{n+2} - \frac{1}{n+1}$$

we can simplify

$$\begin{aligned} S_N &= \sum_{n=0}^N \frac{1}{n+2} - \frac{1}{n+1} \\ &= \left( \frac{1}{0+2} - \frac{1}{0+1} \right) + \left( \frac{1}{1+2} - \frac{1}{1+1} \right) + \dots + \left( \frac{1}{N+2} - \frac{1}{N+1} \right) \end{aligned}$$

Thus we have all the terms cancel except the first part of the Nth term and the second part of the initial term. So

$$\begin{aligned} S_N &= \sum_{n=0}^N \frac{1}{n+2} - \frac{1}{n+1} \\ &= -\frac{1}{0+1} + \frac{1}{N+2} \end{aligned}$$

And to turn this into an infinite series take lets take the limit as N goes to infinity.

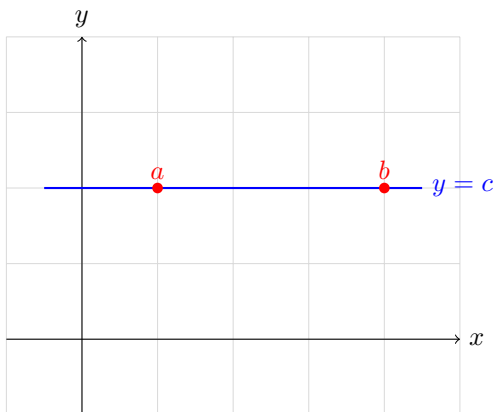
$$\begin{aligned}
S &= \sum_{n=0}^{\infty} \frac{1}{n+2} - \frac{1}{n+1} \\
&= \lim_{N \rightarrow \infty} -\frac{1}{0+1} + \frac{1}{N+2} \\
&= -\frac{1}{1} + \frac{1}{\infty+2} \\
&= -1 + \frac{1}{\infty} \\
&= -1 + 0 \\
S &= -1
\end{aligned}$$

Note we sidestepped convergence. And one of the most important parts of series is convergence. Why did we do this? Because it is really to big a topic for this short review and is not the most important for engineering. Rather, integrals and derivatives are more pivotal to understand. For more information on series I recommend turning to Wikipedia.

Before turning our attention to integrals I want to highlight a skill learned in calculus that is paramount to engineering. Finding tolerances and error bounds! Riddled all through the more rigorous mathematical study of calculus, a subject called Analysis which is finding and bounding functions. In fact it can be said that just as calculus is the math of change, Analysis is the math of bounding that change. So why not go into it? It heavily depends on set theory and quite frankly we do not possess the foundation to effectively discuss this. But for those who are interested I recommend taking the course discrete mathematics and building your knowledge with an analysis text book. Some recommendations are "Understanding Analysis" By Stephen Abbott, or "Analysis with an introduction to proof" by Steven R. Lay.

This will deepen your understanding of how to build mathematical bounds which is a great skill to have when building mathematical models for engineering.

Now, without further adu, lets look at an integral. Specifically the definite integral of a constant function. Let  $c, a, b \in \mathbb{R}$  and consider



$$\int_a^b c dx$$

Then we apply the power rule from the identities sheet

$$\begin{aligned}\int_a^b c dx &= cx \Big|_a^b \\ &= cb - ca \\ &= c(b - a)\end{aligned}$$

now looking from a similar perspective lets calculate the area of this rectangle, which is length times width so as length is  $b - a$  we have that term and width is the height  $c$  we have that term. Thus our area  $A$  is

$$A = c(b - a)$$

so as we can see our area is the integral.

$$A = \int_a^b f(x) dx$$